

SQL Server Lesson 1 Summary

Summary: SQL Server Basics

Lesson 1: Introduction to SQL Server

1. What is SQL Server?

SQL Server is a Relational Database Management System (RDBMS) by Microsoft used to store and manage structured data.

2. Tools to Install:

- SQL Server Engine (Developer Edition recommended)
- SQL Server Management Studio (SSMS)

3. First Commands:

USE TestDB;

```
CREATE TABLE Students (  
    StudentID INT PRIMARY KEY,  
    Name NVARCHAR(100),  
    Age INT,  
    Email NVARCHAR(100)  
);
```

4. Insert and View Data:

```
INSERT INTO Students (StudentID, Name, Age, Email)  
VALUES (1, 'Alice', 21, 'alice@college.edu');
```

```
SELECT * FROM Students;
```

5. Check Table:

```
SELECT name FROM sys.tables;
```

```
SELECT * FROM INFORMATION_SCHEMA.COLUMNS WHERE TABLE_NAME = 'Students';
```

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6. Insert More Data Dynamically:

```
INSERT INTO Students (StudentID, Name, Age, Email)
VALUES (8, 'Hana', 23, LOWER('Hana') + '@college.edu');
```

7. Handling NULL Constraints:

- If column does not allow NULL, either provide a value or alter the column:

```
ALTER TABLE Students ALTER COLUMN Email NVARCHAR(100) NULL;
```

8. Update Email Dynamically:

```
UPDATE Students
SET Email = LOWER(REPLACE(Name, ' ', '')) + '@college.edu'
WHERE Email IS NULL;
```

9. Update Specific Rows:

```
UPDATE Students
SET Email = 'alice@college.edu'
WHERE Name = 'Alice';
```

10. Insert Using SELECT + Expressions:

```
INSERT INTO Students (StudentID, Name, Age, Email)
SELECT 9, 'Ibrahim', 24, LOWER('Ibrahim') + '@college.edu';
```